BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

MATHEMATICS (10th) — (Fresh/Re-appear) — (Version-A) — SECTION – A

Time Allowed: 20 Min | **Maximum Marks:** 15 | **Paper Code:** BN-X-XXIII-I

Note: Note: There are THREE sections in this paper i.e A, B & C. Attempt section A and return it to the superintendent within the given time. Q.No.1: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct fill that circle in front of the question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

Q1. What are the solutions of the equation $x^2 + 7x - 18 = 0$?

(A) 2 or -9
(B) -2 or 9
(C) -2 or -9
(D) 2 or 9

Q2. For what value of K the roots of equation $x^2 - 5x + k = 0$ are equal

(A) 4/25
(B) -4/25
(C) -25/4
(D) 25/4

Q3. The third proportional of x and y is

(A) xy
(B) x/y
(C) y^2/x
(D) x=81

Q4. The most frequent value in the data is called its _____

(A) Mean
(B) Median
(C) Mode
(D) G.M

Q5. The average pocket money of 30 students is Rs 20/-. The total amount in the class is

(A) Rs 20/-
(B) Rs 30/-
(C) Rs 300/-
(D) Rs 600/-

Q6. In the data 5,5,5,5 the standard deviation is

(A) 5
(B) 0
(C) 4
(D) None of these

Q7. Sin(-350°) lies in which quadrant.

(A) 1st
(B) 2nd
(C) 3rd
(D) 4th

Q8. Apollonius was born in

(A) Perga
(B) Istanbul
(C) France
(D) Germany

Q9. Median of 2, 4, 5, 6, 3 is

(A) 2
(B) 3
(C) 4
(D) 5

Q10. Mean proportional of 12, 3 is

(A) 3
(B) 6
(C) 9
(D) 12

Q11. The sum of the roots of equation $ax^2+bx+c = 0$ is

(A) b/a
(B) -b/a
(C) c/a
(D) -c/a

Q12. The nth root of product of n number of values is called

(A) Arithmetic Mean
(B) Geometric Mean
(C) Harmonic Motion
(D) Standard Deviation

Q13. A chord is 5cm from the centre of a circle of radius 13cm. The length of the chord is

(A) 6cm
(B) 12cm
(C) 24cm
(D) 30cm

Q14. If f is a function from A to B, then f is onto function if

(A) Range f = B
(B) Range f ≠ A
(C) Domain f = A
(D) None of these

Q15. The sum of the cube roots of unity is

(A) 1
(B) -1
(C) 0
(D) None of these

MATHEMATICS (10th) — (Fresh/Re-appear) — SECTION –B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 36

Note: Note: Attempt section B & C accordingly.

Q2. (Marks 36) Q.No 2. Attempt any NINE parts of the following. All parts carry equal marks.

i. Solve $3 \cdot 3^{(2x+1)} - 10 \cdot 3^x + 1 = 0$

ii. Show that $(-1 + \sqrt{3}i)^3 + (-1 - \sqrt{3}i)^3 = 16$

iii. The sum of two number is 11 and sum of their square is 65. Find the number.

iv. Volume of gas "V" varies inversely as pressure "p". $P = 300N/m^2$ when $V = 4m^3$ find pressure when $V = 3m^3$

v. Domain of a binary relation $R = \{(x,y) \mid y+1 = 2x^2\}$ is set of N. Find range.

vi. A set of data contains the values as 148, 145, 160, 157, 156 and 160. Show that Mode > Median > Mean

vii. Prove that $\tan\theta + \sec\theta = (1 + \sin\theta) / \cos\theta$

viii. A tree is 50m high. Find the angle of elevation of its top to a point on the ground 100m away from the foot of tree

ix. Resolve into partial fraction $(2x^2+1) / (x-2)^2(x+3)$

x. If $((x+3)^2 + (x-4)^2) / ((x+3)^2 - (x-4)^2) = 13/12$, find the value of x.

xi. Use synthetic division to find the value of k if 2 is a zero of polynomial $2x^4 + x^3 + kx^2 - 8$

xii. If $11 : x - 1 = 2 : 27$, find value of x

SECTION –C

Maximum Marks: 24

Note: Note: Attempt any THREE of the following. All questions carry equal marks.

Q3. The two tangents drawn to a circle from a point outside it, are equal in length.

Q4. Perpendicular from the center of a circle on a chord bisect it.

Q5. The angle in a semi circle is a right angle.

Q6. Inscribe a square of side 10cm in a circle. What will be the size of the radius?